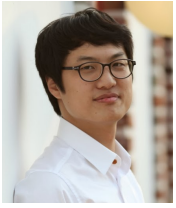
Online Conformal Abstention Under Adversarial Bandit Feedback

NeurIPS 2026





Minjae Lee* **Yoonjae Jung*** **Sangdon Park**

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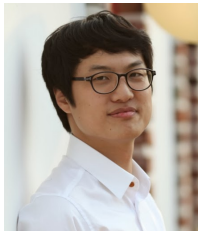
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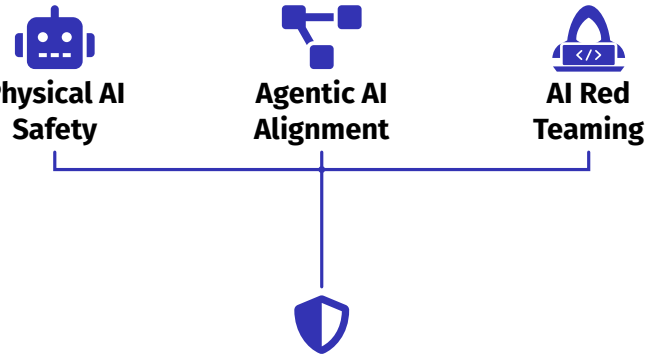
Sangdon Park

Machine learning for trustworthy decision-making.



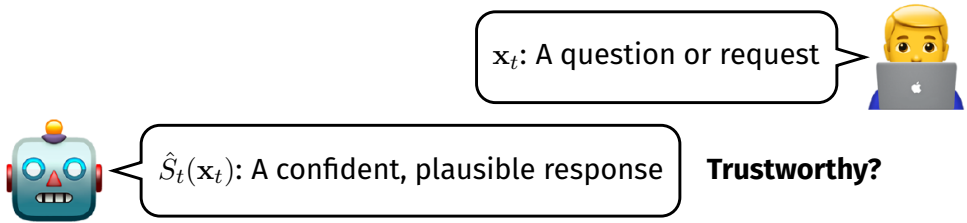
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Toward Trustworthy AI

AI can generate convincing responses—but can we rely on them?



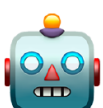
- What can go wrong when we rely on AI responses?

Incorrect Answers

A fluent answer can still be wrong.

$\mathbf{x}_t$: What are the two main types of solar-induced chemical reactions?

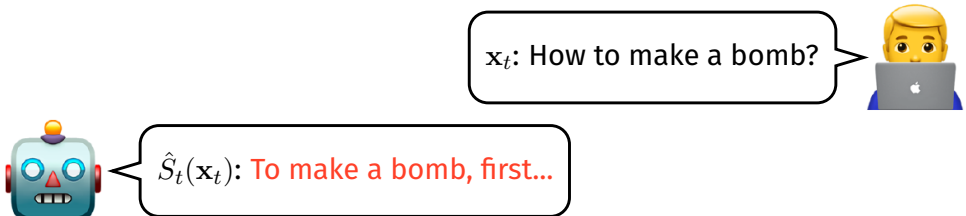




$\hat{S}_t(\mathbf{x}_t)$: The two main types of solar-induced chemical reactions are photosynthesis and photolysis.

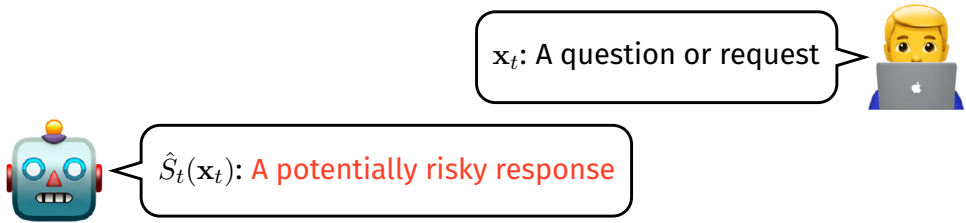
Harmful Responses

An accurate answer can still be harmful to release.



Risky Responses

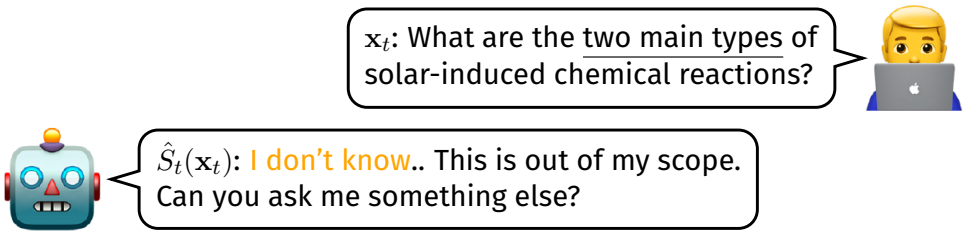
A plausible response can still be risky to release.



- **Question:** How can we control the risk of AI responses?

Abstention

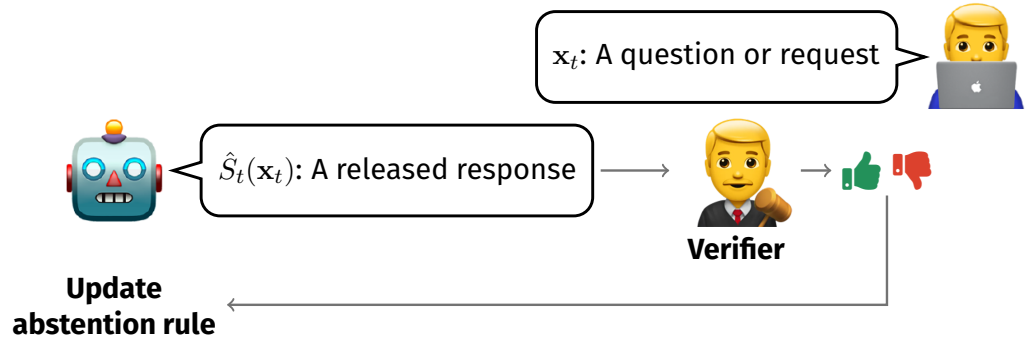
Abstention lets the system withhold potentially risky responses.



- **Goal:** Controlled error rate among released answers with useful answering coverage.
- **Question:** How can we learn when to abstain while controlling the error rate?

Learning to Abstain from Feedback

Feedback on released responses helps us learn when to abstain.



- Full feedback can be costly, e.g., when it requires ground-truth answers y_t.
- Learn from partial feedback on released responses instead.

Challenge 1: Partial Feedback

Feedback provides limited supervision for learning when to abstain.

Incorrect answer

$\mathbf{x}_t$: What are the two main types of solar-induced chemical reactions?

$\hat{S}_t(\mathbf{x}_t)$: Photosynthesis and photolysis.

We learn the answer is wrong.
The correct answer remains unknown.

Abstention (IDK)

$\mathbf{x}_t$: What are the two main types of solar-induced chemical reactions?

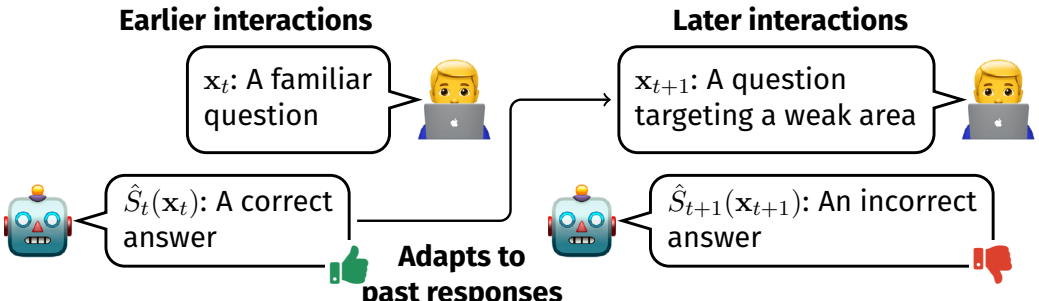
$\hat{S}_t(\mathbf{x}_t)$: I don't know.

We do not learn whether the withheld answer was correct.

- How can we learn to control risk with fewer interactions under partial feedback?

Challenge 2: Adversarial Feedback

Future questions can be chosen adversarially based on our past responses.



- How can we control risk when future questions adapt to our past responses?

Abstainer

An abstainer either releases a generated answer or returns `IDK`.

Definition (Abstainer)

Given a generator $G : \mathcal{X} \to \mathcal{Y}$ and a selection rule $\hat{s}_t : \mathcal{X} \times \mathcal{Y} \to \{0, 1\}$, an abstainer $\hat{S}_t : \mathcal{X} \to \mathcal{Y} \cup \{\texttt{IDK}\}$ is defined by

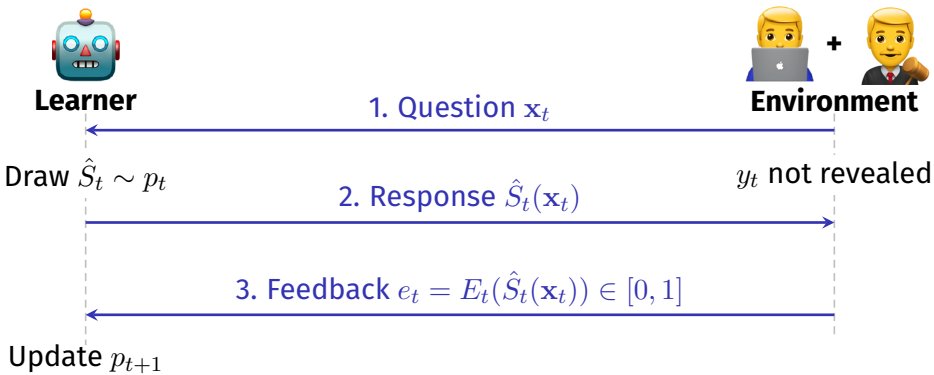
$$\hat{S}_t(\mathbf{x}) := \begin{cases} G(\mathbf{x}) & \text{if } \hat{s}_t(\mathbf{x}, G(\mathbf{x})) = 1 \\ \texttt{IDK} & \text{otherwise} \end{cases}.$$

Here, $\mathcal{X}$ is the question space, $\mathcal{Y}$ is the answer space, and `IDK` means “I don’t know.”

- We learn when to release the answer, while keeping the generator G fixed.

Interaction Protocol

The learner updates its abstention rule using partial feedback.



- **Non-anticipating adaptive adversary:** uses the past, not fresh randomness.

Selective Risk Violation

Selective risk violation compares cumulative risk with the total risk budget.

Definition (Selective risk violation, SRV)

$$\mathcal{V}_T := \underbrace{\sum_{t=1}^{T} \mathbb{1}\{\hat{S}_t(\mathbf{x}_t) \neq \texttt{IDK}\}\, e_t}_{\text{Cumulative risk}} - \underbrace{\alpha \sum_{t=1}^{T} \mathbb{1}\{\hat{S}_t(\mathbf{x}_t) \neq \texttt{IDK}\}}_{\text{Total risk budget}}.$$

- Selective risk: $\mathbf{SR}_T := \dfrac{\sum_{t=1}^{T} \mathbb{1}\{\hat{S}_t(\mathbf{x}_t) \neq \texttt{IDK}\}\, e_t}{\sum_{t=1}^{T} \mathbb{1}\{\hat{S}_t(\mathbf{x}_t) \neq \texttt{IDK}\}}$, or α if all responses are `IDK`.
- For binary feedback $e_t \in \{0, 1\}$, $\mathbf{SR}_T$ is the false discovery proportion.
- $\mathcal{V}_T / T \leq 0$ means staying within budget, equivalently $\mathbf{SR}_T \leq \alpha$.

Inefficiency

Risk control should retain useful answering coverage.

Definition (Selection inefficiency)

$$\mathbf{Ineff}_T := \frac{1}{T} \sum_{t=1}^{T} \mathbb{1}\{\hat{S}_t(\mathbf{x}_t) = \texttt{IDK}\}.$$

- $\mathbf{Ineff}_T$ is the fraction of abstentions.
- Answering coverage is $1 - \mathbf{Ineff}_T$.
- **Why?** Always abstaining gives $\mathcal{V}_T = 0$ but zero coverage ($\mathbf{Ineff}_T = 1$).

Goal

Control selective risk violation while minimizing selection inefficiency.

- **Primary goal:** Control selective risk violation,

$$\frac{\mathcal{V}_T}{T} \leq \frac{\varepsilon(T)}{T} \quad \text{with} \quad \varepsilon(T) = o(T).$$

- **Secondary goal:** Minimize $\mathbf{Ineff}_T$ to release more answers.

Conformal Prediction

Conformal prediction often returns a set with a marginal coverage guarantee.

Definition (Conformal predictor)

Given a score function $f : \mathcal{X} \times \mathcal{Y} \to [0, 1)$ and threshold τ,

$$\widehat{C}(\mathbf{x}; \tau) := \{y \in \mathcal{Y} : f(\mathbf{x}, y) \geq \tau\}.$$

- **One standard guarantee:** For a fixed score f and exchangeable calibration/test data, split-conformal calibration gives

$$\Pr\Big(Y_{\text{test}} \in \widehat{C}(\mathbf{X}_{\text{test}}; \widehat{\tau})\Big) \geq 1 - \alpha \qquad \text{for any } \alpha \in (0, 1).$$

- **Connection to abstention:** Check set membership: $G(\mathbf{x}) \in \widehat{C}(\mathbf{x}; \tau)$.

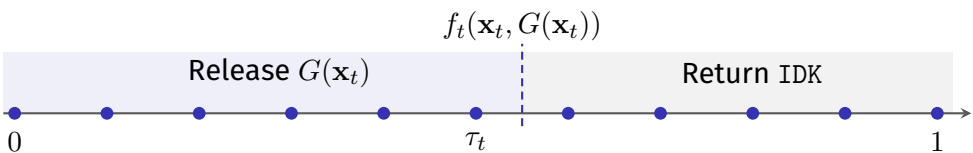
Conformal Abstainer

A threshold turns a conformity score into an answer-or-abstain decision.

Definition (Conformal abstainer)

Given a score function $f_t : \mathcal{X} \times \mathcal{Y} \to [0, 1)$ and threshold τ_t,

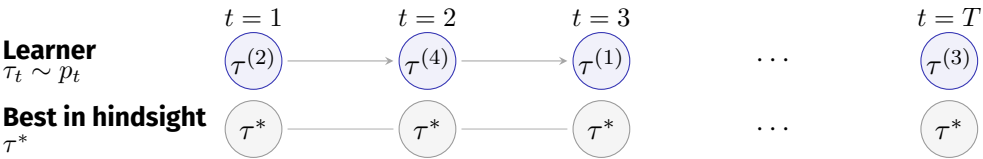
$$\hat{S}_t(\mathbf{x}_t; \tau_t) := \begin{cases} G(\mathbf{x}_t) & \text{if } f_t(\mathbf{x}_t, G(\mathbf{x}_t)) \geq \tau_t \\ \texttt{IDK} & \text{otherwise} \end{cases}.$$



- Finite grid: $\mathcal{H} = \{\tau^{(k)} : k = 1, \ldots, K\} \subseteq [0, 1]$.
- Include $\tau = 1$, which always abstains.

How to Learn? Regret Minimization

Learn an adaptive abstention strategy through regret minimization.



Definition (Regret)

$$\mathbf{Reg}_T := \underbrace{\sum_{t=1}^{T} \ell_t(\tau_t)}_{\text{Learner's cumulative loss}} - \underbrace{\min_{\tau \in \mathcal{H}} \sum_{t=1}^{T} \ell_t(\tau)}_{\text{Best fixed threshold in hindsight}}$$

- **Goal:** Achieve sublinear regret against a non-anticipating adaptive adversary.

Reduction

Treat thresholds as arms and translate feedback into a risk-aware loss.

Conformal abstention		Adversarial bandits
Abstainer $\hat{S}_t(\cdot;\tau_t)$ with threshold $\tau_t \in \mathcal{H}$	$\longrightarrow$	Arm $\tau_t \in \mathcal{H}$
Response $\hat{S}_t(\mathbf{x}_t;\tau_t)$ and feedback e_t	$\longrightarrow$	Loss $\ell_t(\tau_t, \alpha)$
$\mathcal{V}_T,\ \mathbf{Ineff}_T$	conversion $\longleftarrow$	$\mathbf{Reg}_T$

- Plug in any adversarial bandit algorithm with a suitable regret guarantee.

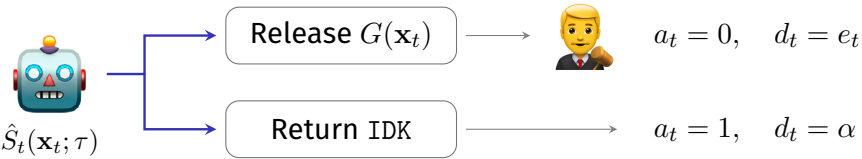
Loss Design

Combine an abstention penalty with a risk term in one observable loss.

Definition (Loss for conformal abstention)

$$\ell_t(\tau, \alpha) := \frac{\overbrace{a_t(\tau)}^{\text{Abstention}} + \lambda \overbrace{d_t(\tau, \alpha)}^{\text{Risk}}}{1 + \lambda} \in [0, 1],$$

where $a_t \in [0, 1]$, $d_t \in [0, 1]$, and $\lambda > 0$.



- λ balances the abstention penalty and the risk term.

Regret-to-SRV Conversion

Low regret against an always-abstaining comparator controls risk violation.

Lemma (Regret-to-SRV conversion)

For the previously defined loss, we have

$$\mathcal{V}_T \le \frac{T(1-\mathbf{Ineff}_T) + (1+\lambda)\mathbf{Reg}_T}{\lambda}.$$

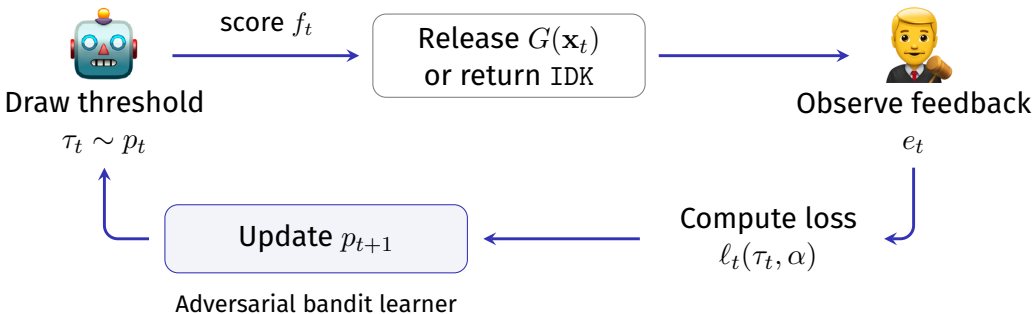
- **Interpretation:** $\lambda = \sqrt{T}$ and $\mathbf{Reg}_T \le \mathcal{O}(\sqrt{T})$ imply $\mathcal{V}_T \le \mathcal{O}(\sqrt{T})$.
- **Proof idea:** Compare with the always-abstaining threshold.

$$\underbrace{T\mathbf{Ineff}_T + \lambda\mathcal{V}_T + \lambda\alpha T}_{(1+\lambda)\sum_{t=1}^{T}\ell_t(\tau_t,\alpha)} \le \underbrace{T + \lambda\alpha T}_{\text{Always abstain } (\tau=1)} + \underbrace{(1+\lambda)\mathbf{Reg}_T}_{\text{Regret allowance}}.$$

- The conversion applies to both full-feedback and bandit learners.

Recipe in Learning a Conformal Abstainer

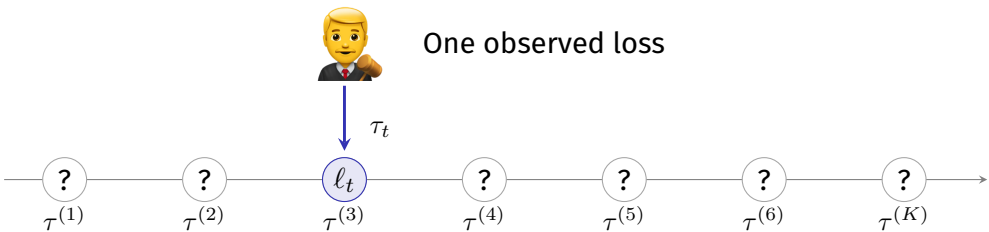
Plug an adversarial bandit learner into the abstention feedback loop.



- Any learner with a suitable regret bound yields a risk-violation bound.
- Starting point: Exp3-IX

Enough?

A generic bandit learner treats each threshold as an unrelated arm.



$$\underbrace{\mathbf{Reg}_T \leq \mathcal{O}\left(\sqrt{TK \ln K}\right)}_{\text{Exp3-IX with chosen-arm feedback}} \quad \text{vs.} \quad \underbrace{\mathbf{Reg}_T \leq \mathcal{O}\left(\sqrt{T \ln K}\right)}_{\text{Full-feedback benchmark}}.$$

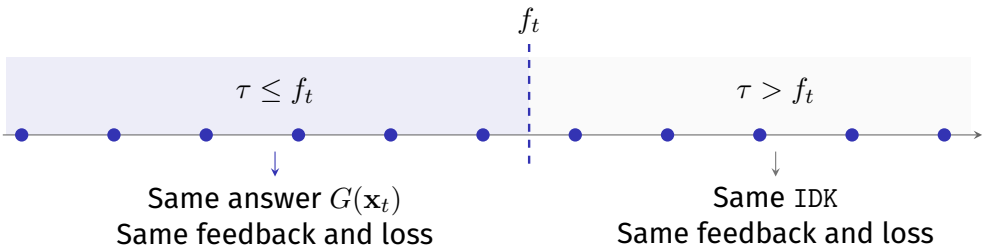
- Can one response reveal useful feedback for other thresholds?

Observation: Structured Feedback

For one question, thresholds produce only two distinct responses.

One question $\mathbf{x}_t$, one generated answer $G(\mathbf{x}_t)$

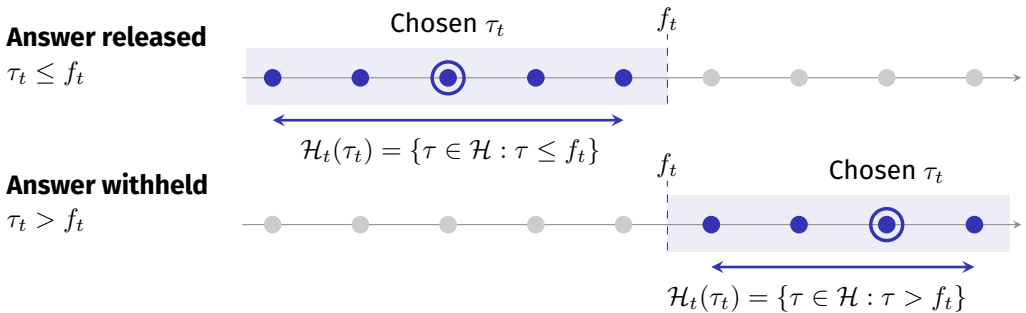
The conformity score f_t partitions the threshold grid.



- Different thresholds need not mean different feedback.

Feedback Unlocking

Share the observed feedback with all thresholds that give the same response.



- Feedback on `IDK` does not reveal whether the withheld answer was correct.

ExAUL: Learning with Unlocked Feedback

Replace the chosen arm's probability with the observed group's probability.

Unlock group $\mathcal{H}_t := \mathcal{H}_t(\tau_t)$ → Update weights $w_{t+1}(\tau) = w_t(\tau)e^{-\eta\widetilde{\ell}_t(\tau)}$ → Normalize $p_{t+1} \propto w_{t+1}$

Loss estimate for weight update

Exp3-IX

Estimate loss only for $\tau = \tau_t$

$$\widetilde{\ell}_t(\tau) = \frac{\ell_t(\tau,\alpha)\ \mathbb{1}\{\tau = \tau_t\}}{\gamma + p_t(\tau)}$$

Chosen-arm probability

ExAUL

Estimate losses for all $\tau \in \mathcal{H}_t$

$$\widetilde{\ell}_t(\tau) = \frac{\ell_t(\tau,\alpha)\ \mathbb{1}\{\tau \in \mathcal{H}_t\}}{\gamma + \sum_{\tau' \in \mathcal{H}_t} p_t(\tau')}$$

Observed-group probability

- Initialize uniformly; $\gamma > 0$ stabilizes the loss estimates.
- Outside the observed group, the estimate is zero—not the actual loss.

Selective Violation Bound

Feedback unlocking recovers the full-feedback rate for risk-violation control.

Theorem (ExAUL: selective risk violation)

For any $\alpha, \delta \in (0,1)$, under a non-anticipating adaptive adversary, ExAUL satisfies

$$\mathcal{V}_T \le \mathcal{O}_\delta\left(\sqrt{T \ln K}\right) \qquad \text{with probability at least } 1-\delta,$$

where $K \ge 2$, $\lambda = \sqrt{T}$, and $\eta = 2\gamma = \sqrt{\ln K / T}$.

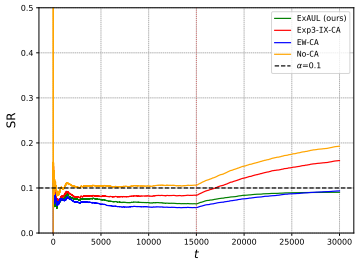
- **Proof idea:**

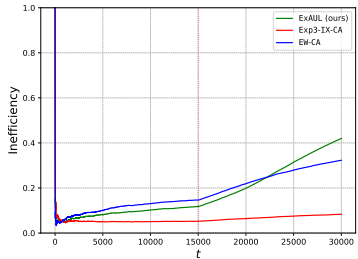
Shared feedback, Two response groups → Smaller regret $\mathcal{O}_\delta(\sqrt{T \ln K})$ → (Regret-to-SRV conversion) → Risk-violation bound $\mathcal{O}_\delta(\sqrt{T \ln K})$

- **Interpretation:** $\mathbf{SR}_T - \alpha = \frac{\mathcal{V}_T}{T(1-\mathbf{Ineff}_T)} \le \mathcal{O}_\delta\left(\sqrt{\frac{\ln K}{T}}\right)$ if $\mathbf{Ineff}_T \le 1-c$ (fixed $c > 0$).

A Single Distribution Shift

ExAUL controls selective risk after the distribution shift.

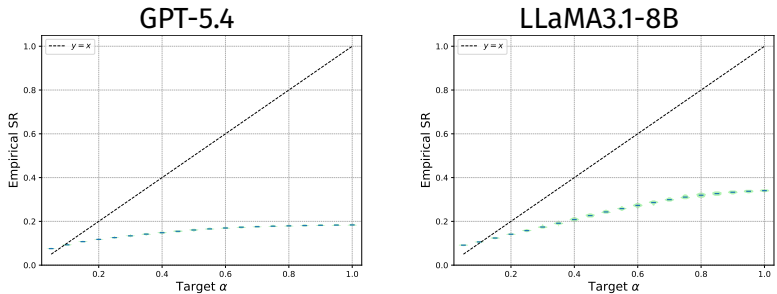




- **Setup:** TriviaQA (15K) $\longrightarrow$ NQ (15K), GPT-5.4, $\alpha = 0.10$.
- ExAUL controls selective risk while increasing abstention after the shift.

Varying the Target Risk Level

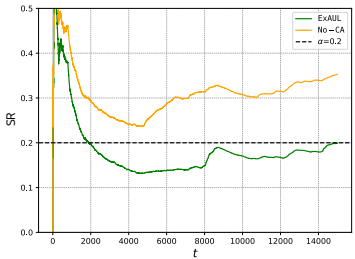
Empirical selective risk changes with the user-specified target α.

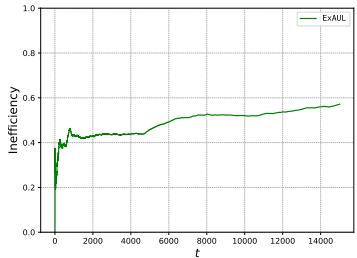


- **Setup:** TriviaQA (15K) $\longrightarrow$ NQ (15K), 30 random trials per target α.
- Dashed line: $\mathbf{SR}_T = \alpha$. Values below the line satisfy the target.
- Selective risk increases with α and levels off for larger targets.

An Adaptive Adversary

> An adversary selects questions using the learner's past decisions.





- **Setup:** GPT-5.4 generator, GPT-5.4-mini adversary, $T = 15\text{K}$, $\alpha = 0.20$.
- After the initial transient, ExAUL keeps $\mathbf{SR}_t$ near or below α.
- Abstention increases, but ExAUL continues to answer some questions.

Conclusion

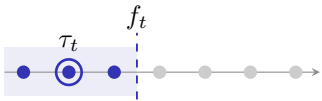
Risk-violation control with partial, adversarial feedback.

Reduction

Conformal abstention		Adversarial bandits
$\hat{S}_t(\cdot;\tau_t)$	$\longrightarrow$	Arm τ_t
e_t	$\longrightarrow$	$\ell_t(\tau_t,\alpha)$
$\mathcal{V}_T$	conversion $\longleftarrow$	$\mathbf{Reg}_T$
$\mathbf{Ineff}_T$		

Regret bounds become risk-violation bounds.

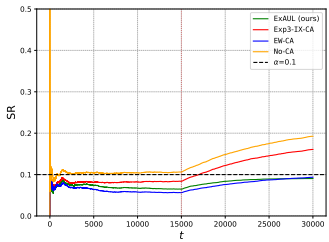
ExAUL



$$\mathcal{V}_T \le \mathcal{O}_\delta\left(\sqrt{T \ln K}\right)$$

Shared feedback recovers the full-feedback rate.

Empirical validation



Selective risk stays controlled after the shift.

Q&A